NeST-Com: Fixed-Budget Next-Scale Token Communication

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Abstract

Next-scale residual tokens support source-referenced perceptual reconstruction from coarse descriptions, but whole-scale transmission can underfill fixed digital frames. NeST-Com fits independently terminated token-wise prefixes to the available capacity and uses received boundary tokens for residual correction and finer-scale conditioning. At 1024 symbols and 10 dB, source-bit utilization rises from 77.70% to 99.79%, reducing DISTS and DreamSim by 8.7% and 20.5% relative to shared-model whole-scale transmission. Equal-information-budget tests favor NeST-Com over adapted Ada-TokenCom on both distances at the three smallest tested budgets. At 1024 symbols and 10 dB, transmitter-plus-receiver processing takes 0.66 s versus 15.42 s for Ada on matched hardware.

Index Terms

Image communication, next-scale tokens, fixed symbol budget, perceptual reconstruction.

I. Introduction

A hard per-image symbol budget must accommodate source information, signaling and protection. We study source-referenced perceptual reconstruction under this constraint. Transmitted information anchors reconstruction to the source, while a learned prior predicts untransmitted

content. The representation's organization and use of payload capacity therefore affect reconstruction quality.

Next-scale tokenization provides complete coarse maps over the full latent grid, followed by residual refinements [1]. Within a scale, token probabilities share the completed coarse context, enabling scale-wise evaluation and completion. ARPC uses next-scale prediction for progressive compression, transmitting coarse scales and generating the remaining scales [2]; Ada-TokenCom combines next-token prefix coding, tail generation and wireless adaptation [3]. ResARC generates quantization residuals and transmits a learned code for suffix-generation residuals [4]. Here, the transmitted refinements are source tokens in the existing next-scale representation.

Progressive transmission matches source descriptions to channel resources [5]. A full next-scale map may exceed remaining frame capacity although much of it would fit. Entropy-coded lengths also vary across images. Exploiting this capacity requires independently decodable within-scale endpoints, actual-length fitting and reconstruction that uses received boundary tokens in both the residual representation and finer-scale context.

NeST-Com adapts next-scale source descriptions to a fixed per-image digital frame. It fits independently terminated within-scale prefixes using their actual coded lengths and uses received boundary tokens in residual reconstruction and finer-scale conditioning. Shared-model comparisons and matched controls establish how the additional source information improves reconstruction. Equal-information-budget comparisons identify a favorable tight-budget perceptual regime, while measured processing costs quantify the tradeoff between endpoint granularity and reconstruction quality.

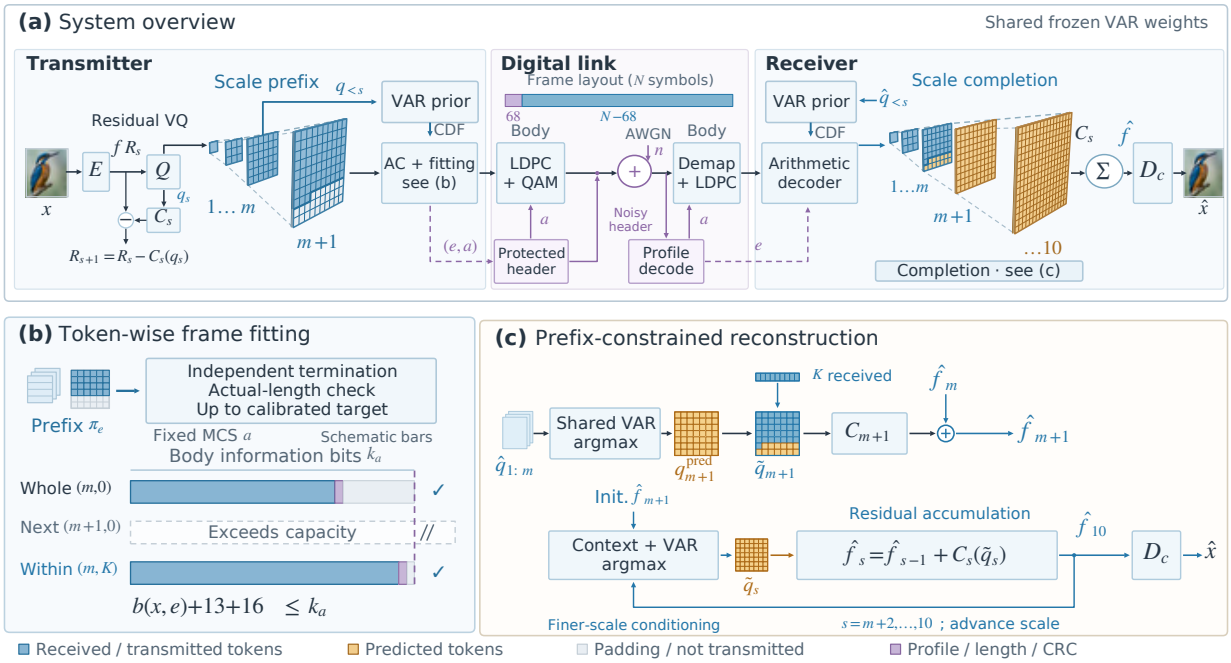


Fig. 1. NeST-Com. (a) The selected endpoint and fixed MCS determine the protected profile, combined with the coded body before the channel. Both terminals reconstruct coding probabilities from shared VAR and known coarse tokens. (b) Independently terminated prefixes undergo actual-length fitting up to the calibrated target; bars are schematic. (c) Received tokens overwrite boundary predictions. Each remaining scale recursively updates the accumulated state and constructs the next context before final decoding. Unreceived boundary positions retain their coarse-conditioned predictions.

II. Proposed NeST-Com Framework

A. Next-Scale Representation and Digital Frame

We transmit $x \in [0,1]^{256\times256\times3}$ using N complex symbols over AWGN,

$$y = s + n, \quad n \sim \mathcal{CN}(0, \sigma^2 I_N), \quad \mathrm{SNR} = 10\log_{10}(E_s/\sigma^2), \tag{1}$$

with nominal $E_s = 2$. TX uses the frozen policy for its design SNR; RX uses the actual noise variance for LLRs. Digital constellations have average energy two without per-frame renormalization. A 12-bit profile ID, CRC16, and terminated rate-1/2 convolutional code occupy 68 rotated-BPSK symbols $\{\pm(1+j)\}$. The profile identifies the endpoint and MCS. The remaining

$N-68$ symbols carry the LDPC-coded body, with public frame-dependent scrambling after rate matching and inverse scrambling before decoding. Continuous JSCC data are normalized per image to energy two; required side information is paid within N.

The frozen VAR tokenizer produces ten residual token maps q_s, with side lengths $(1, 2, 3, 4, 5, 6, 8, 10, 13, 16)$ and a 4096-entry codebook [1]. Write L_s for the number of positions and C_s for codebook lookup, resizing to the full latent grid, and the scale-dependent residual transform. The latent sum $\sum_s C_s(q_s)$ progressively refines a full-field description. Decoder D_c is adapted on 20 000 ImageNet training images using MSE plus 0.1 LPIPS, then frozen. The continuous/quantized/prefix-completed training mixture is specified in Supplement S1. The null-class VAR-d16 prior factorizes as

$$p_\theta(q_s \mid q_{<s}) = \prod_{i=1}^{L_s} p_\theta(q_{s,i} \mid q_{<s}). \tag{2}$$

Both terminals reconstruct identical integer CDFs, with total frequency 2^{24}, for arithmetic coding [6]; no unreceived same-scale token is needed.

B. Within-Scale Endpoints and Actual-Length Fitting

An endpoint $e = (m, K)$ retains complete coarse maps and a raster-ordered boundary prefix:

$$\pi_e = (q_1, \ldots, q_m, q_{m+1,1:K}). \tag{3}$$

The boundary prefix is empty for $K = 0$, including $e = (10, 0)$. The fixed raster order specifies received positions without a mask. WHOLE permits $(m, 0)$ for $m = 4, \ldots, 10$. NeST permits every integer $K = 0, \ldots, L_{m+1}-1$ for $m = 4, \ldots, 9$, together with $(10, 0)$: 651 distinct endpoints from 30 through 680 retained tokens. The paid 12-bit profile directory covers the applicable endpoint/MCS combinations. Quarter and Eighth provide coarser within-scale configurations; Token is the main configuration.

For MCS $a = (M, r)$, the coded body has $n_a = (N - 68) \log_2 M$ bits and information length $k_a = \lfloor r n_a \rfloor$. Its independently terminated arithmetic length $b(x, e)$ must satisfy

$$b(x, e) + 29 \leq k_a, \qquad \eta(x, e) = \frac{b(x, e)}{k_a - 29}. \tag{4}$$

The 29 bits comprise a 13-bit length and CRC16. Unused information positions are zero-padded before 5G NR LDPC coding, rate matching [7], and QAM. Thus η measures source-bit capacity utilization after the length/CRC overhead.

Offline calibration fixes a target and MCS for each budget/SNR by DINOv2 source/reconstruction cosine [8]. NeST retains these targets/MCSs while refining endpoints; Supplement S1 specifies the family-specific candidate pools. A shared scale-wise probability and incremental coding traversal supplies candidates, each independently terminated rather than cut from a longer stream. With MCS fixed, TX selects the feasible endpoint with the most tokens up to the calibrated target. Actual terminated lengths determine feasibility without assuming length monotonicity. The timed implementation scans all allowed candidates. If no legal endpoint fits, TX sends a paid NULL profile with a two-bit zero payload; valid reception yields a gray image. Supplement S1 specifies the directory and fallback rules.

C. Prefix-Constrained Reconstruction

Invalid profile CRC/catalogue, body CRC, declared length, known padding, or arithmetic parsing yields a fixed gray image, as does valid NULL reception. Otherwise, RX decodes the actual prefix using the shared CDFs. At $e = (10, 0)$, all maps are received and directly decoded. Otherwise, RX predicts the boundary map by argmax conditioned on complete received maps,

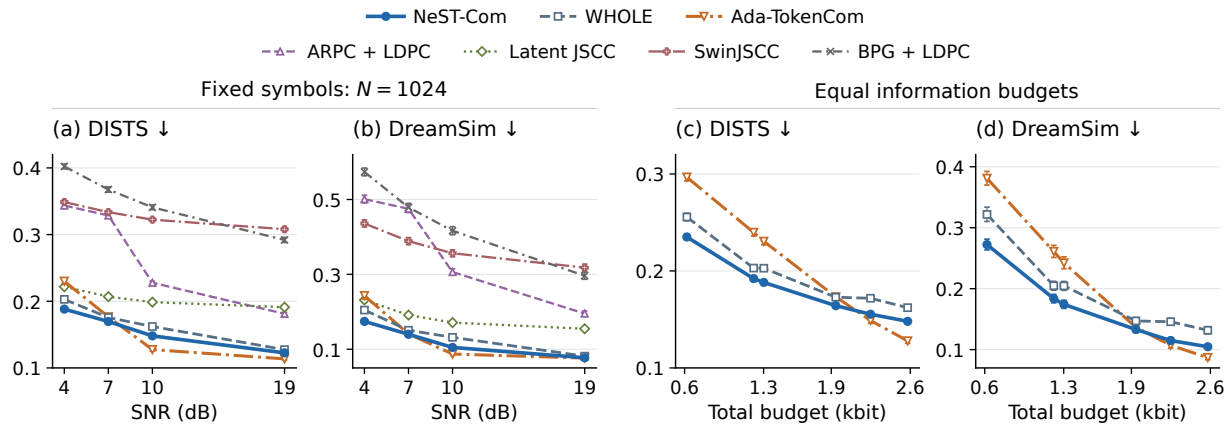


Fig. 2. Source-referenced reconstruction on 500 ImageNet images. (a,b) Seven systems at $N = 1024$; three channel outcomes per source include gray failures. (c,d) NeST-Com, WHOLE and Ada-TokenCom at equal, error-free information budgets, including 57 shared protocol bits and Ada-TokenCom's additional 10-bit class; WHOLE/NeST share ceiling $(10, 0)$. Points and error bars are equal-source means and pointwise 95% source-bootstrap confidence intervals after within-source averaging. Ada-TokenCom is the predicted-class, fixed-frame adaptation. Lower values are favorable.

then overwrites known positions [1]:

$$\tilde{q}_{m+1,i} = \begin{cases} \hat{q}_{m+1,i}, & i \le K, \\ q^{\text{pred}}_{m+1,i}, & i > K. \end{cases} \tag{5}$$

Received boundary tokens serve two roles: they correct the residual representation and, through the merged boundary map, condition finer-scale prediction. Unreceived boundary positions retain their original coarse-conditioned prediction; there is no same-scale prediction loop. Complete received maps remain unchanged. After generating the finer maps,

$$\hat{x} = D_c\left(\sum_{s=1}^{10} C_s(\tilde{q}_s)\right). \tag{6}$$

At the last scale, overwrite only changes the residual contribution.

III. Experimental Results

A. Experimental Setup

The 500 ImageNet validation sources [9] are disjoint from the 1000-image calibration set and use fixed 256×256 preprocessing. We evaluate $N = 1024$ at 4/7/10/19 dB and $N = 512$ at 4/10/19 dB. Three channel realizations are averaged within each source, retaining gray failures, before equal-source means and direct paired 95% source-bootstrap intervals (10 000 resamples). Controls use a fixed 100-image subset with source identities specified in Supplement S1. Main systems use their prescribed independent noise streams; the matched-grid control shares noise.

Latent JSCC maps tokenizer latents to symbols with convolutional layers and shares D_c, using MSE, LPIPS and latent consistency (100k updates). SwinJSCC [10] uses RGB MSE (200k updates, minimum calibration-MSE selection). Both train on 20 000 images for 256×256 and $N = 1024$. Digital baselines include BPG [11] and ARPC [2] with LDPC/QAM. The predicted-class, fixed-frame Ada-TokenCom adaptation [3] retains LlamaGen-L/VQ16, prefix arithmetic coding and sampled completion (CFG=4, $T = 1$, top-$k = 2000$, top-$p = 1$). ConvNeXt predicts its transmitted 10-bit class; ARPC captions and all required fields are paid. The comparison uses per-image frames without Ada's long-term controller or ARQ.

DISTS [12] and DreamSim [13] measure source-referenced perceptual reconstruction; PSNR, LPIPS-Alex [14] and CLIP-I [15] complete the evaluation. Metrics use native float RGB: 256×256 for DISTS and the fixed 224×224 bicubic-antialias route for DreamSim/CLIP. Supplement S1 provides training, candidate pools, numerical routes, integer coding and reference vectors.

Processing uses 32 sources, two warmups and three measurements per source, batch one, one RTX PRO 6000 and two CPU cores. TX spans in-memory RGB to symbols; RX spans symbols

to RGB. Fresh probability evaluation, source coding, link processing and generation are included; loading, disk I/O and scoring are excluded. BPG is CPU-only on the same host/quota.

TABLE I

WITHIN-SCALE TRANSMISSION. (A) SHARED-MODEL NEST–WHOLE DIFFERENCES. (B) MATCHED ENDPOINT GRANULARITY. (C) ORDERED RECEIVER INTERVENTIONS. BRACKETS ARE SOURCE-PAIRED 95% INTERVALS; SAMPLE SIZES ARE SPECIFIED FOR EACH BLOCK.

A. 500 sources, $N = 1024$, family-specific policies

SNR	Util. W→N (%)	ΔDISTS	ΔDreamSim
4	73.81→99.56	-0.0146 [-0.0157, -0.0136]	-0.0303 [-0.0333, -0.0273]
7	83.19→99.75	-0.0060 [-0.0097, -0.0021]	-0.0108 [-0.0157, -0.0057]
10	77.70→99.79	-0.0141 [-0.0153, -0.0128]	-0.0270 [-0.0299, -0.0241]
19	85.24→99.83	-0.0048 [-0.0053, -0.0043]	-0.0043 [-0.0051, -0.0036]

B. 100 quality sources; 32 timing sources; ceiling $(8, 126)$

Grid	Util. (%)	DISTS	DreamSim	TX (ms)
Whole	78.13	0.1650	0.1387	183.5
Quarter	95.19	0.1535	0.1143	214.1
Eighth	97.44	0.1516	0.1137	217.2
Token	99.85	0.1505	0.1115	248.4

C. 100 sources, 10 dB, same received prefixes

Intervention	ΔPSNR (dB)	ΔDISTS	ΔDreamSim
B0–A0	+0.287 [0.240, 0.335]	-0.0062 [-0.0077, -0.0048]	-0.0113 [-0.0147, -0.0083]
C0–B0	+0.537 [0.404, 0.682]	-0.0083 [-0.0102, -0.0065]	-0.0159 [-0.0205, -0.0116]

A0: prior completion without received boundary tokens. B0: A0 plus boundary residual correction. C0: B0 plus updated finer-scale conditioning.

B. Within-Scale Fitting and Granularity

Table I(A) compares NeST with WHOLE using shared tokenizer, prior, coder and receiver under the frozen family policies. Across seven budget/SNR settings, all five metrics favor NeST in the paired intervals. At $N = 1024$, 10 dB, source-bit capacity utilization rises from 77.70% to 99.79%, DISTS falls from 0.1621 to 0.1481, and DreamSim from 0.1317 to 0.1048. PSNR gains span 0.45–0.88 dB over the four $N = 1024$ points. DISTS improves for 83.4% of sources and DreamSim for 83.0%; both improve for 72.2% (Supplement S2).

Block B fixes model, 16-QAM rate 2/3, ceiling $(8, 126)$ and noise. Within-scale grids convert unused capacity into better reconstruction. Eighth retains approximately 92% of the Whole-to-Token gain in PSNR, DISTS and DreamSim. Token raises utilization from 97.44% to 99.85% and adds $0.066\,[0.034, 0.101]$ dB PSNR, with DISTS/DreamSim differences $-0.0011\,[-0.0020, -0.0002]$/$-0.0022\,[-0.0042, -0.0001]$, at 31.1 ms additional TX processing. Eighth thus offers a lower-cost option; Token retains the most feasible source tokens.

Block C uses the same actual NeST prefixes for ordered interventions. A0 completes the boundary and finer maps without using received boundary tokens. B0 adds their residual correction while retaining A0's finer-scale conditioning; C0 also updates that conditioning. At 10 dB, B0–A0 improves PSNR by 0.287 dB and reduces DreamSim by 0.0113; C0–B0 adds 0.537 dB and a 0.0159 reduction. Paired intervals favor both stages for these metrics and DISTS. Boundary tokens therefore contribute through residual correction and subsequent prediction. Supplement S3 includes 7 dB and the final-scale case, where no finer-scale propagation remains.

C. Perceptual Reconstruction and Processing Cost

Fig. 2(a,b) compares seven complete systems under the same per-image symbol constraint, capturing their combined representation, coding and receiver choices. At all four $N = 1024$



Fig. 3. Selected reconstruction examples at $N = 1024$ and 10 dB. Columns show source, WHOLE, SwinJSCC, the predicted-class fixed-frame Ada-TokenCom adaptation, and NeST-Com. Values are full-image DISTS↓ / DreamSim↓ for each method's first prescribed channel realization. Supplement S6 gives source identities, selection details and additional examples.

SNRs, NeST has lower DISTS and DreamSim than ARPC, SwinJSCC, Latent JSCC and BPG: all 32 paired intervals exclude zero. SwinJSCC and BPG have higher PSNR throughout, showing a pixel-fidelity/perceptual-similarity tradeoff at the same symbol budget.

Against Ada, NeST improves both distances at 4 dB and DISTS at 7 dB; at 7 dB, the DreamSim difference interval crosses zero. Ada yields lower distances at 10 dB and lower DISTS at 19 dB, where DreamSim again crosses zero. NeST has higher PSNR throughout. Table II and Supplement S2 report auxiliary metrics; Supplement S2 also lists rejection counts.

Fig. 2(c,d) compares the three token systems at equal information budgets without channel errors, separating their source-rate behavior from link reliability while retaining their model differences. Six totals from 620 to 2577 bits pay 57 shared overhead bits and Ada's additional 10-bit class. WHOLE/NeST share ceiling $(10, 0)$; Ada permits 256 tokens and averages three generation repeats within source. NeST improves all five metrics over WHOLE at every budget.

TABLE II

AUXILIARY QUALITY AND PROCESSING COSTS AT $N = 1024$, 10 DB. QUALITY USES 500 SOURCES; TIMING USES 32 SOURCES, ONE RTX PRO 6000, TWO CPU CORES, BATCH ONE, AND COMPLETE TX/RX PROCESSING BOUNDARIES. BPG USES CPU ONLY.

Method	PSNR (dB)↑	LPIPS ↓	CLIP-I ↑	TX (ms)	RX (ms)
WHOLE	19.31	0.199	0.858	213.8	369.9
NeST-Com	20.19	0.166	0.877	248.4	411.6
Ada-TokenCom	19.56	0.167	0.884	7795.6	7626.1
Latent JSCC	20.80	0.173	0.789	15.4	24.1
SwinJSCC	23.03	0.469	0.669	20.5	17.6
BPG + LDPC	22.54	0.589	0.635	2442.0	31.7
ARPC + LDPC	16.47	0.341	0.713	661.7	797.0

DISTS favors NeST over Ada at the four smallest budgets, and DreamSim at the first three; its 1940-bit interval crosses zero. Ada achieves lower distances at 2248/2577 bits. Increasing 1212 to 1302 bits adds only 3.1 source bits to WHOLE but 89.9 to NeST; DreamSim falls from 0.1837 to 0.1743 for NeST while WHOLE remains near 0.2047.

Ada's completion rule also affects quality. On 100 fixed sources with identical received prefixes and model components, replacing sampling by argmax increases DreamSim by $0.0450\,[0.0282, 0.0621]$, $0.0345\,[0.0232, 0.0461]$ and $0.00358\,[0.00101, 0.00601]$ at N/SNR=512/4, 512/10 and 1024/10, respectively (Supplement S3).

Fig. 3 illustrates the truck's rear markings and foreground contours in the second image, which NeST preserves more clearly than WHOLE. In these two selected 10-dB cases, NeST has lower DISTS and DreamSim than the three displayed comparators, while SwinJSCC has higher

PSNR. Supplement S6 gives source identities, the selection rule, broader method comparisons, and median-rank and degradation examples. At $N = 1024$, 10 dB, NeST requires 0.660 s TX+RX versus 15.422 s for the evaluated Ada implementation, a ratio of 23.4. WHOLE requires 0.584 s. SwinJSCC and Latent JSCC are substantially faster (Table II); Supplement S7 gives the complete cost breakdown.

D. Reduced Budget and Transfer to Kodak

TABLE III

REDUCED-BUDGET RECONSTRUCTION ON 500 SOURCES AT $N = 512$. MEANS INCLUDE ALL FAILURES; PSNR DIFFERENCES USE DIRECT SOURCE-PAIRED 95% INTERVALS. ALL FIVE METRICS AND PAIRED COMPARISONS ARE IN SUPPLEMENT S2.

SNR / method	PSNR↑	DISTS↓	DreamSim↓
4 / WHOLE	15.11	0.2580	0.3247
4 / NeST-Com	15.61	0.2363	0.2740
4 / Ada-TokenCom	11.32	0.2996	0.3847
NeST–WHOLE ΔPSNR: +0.504 [0.440, 0.573] dB			
10 / WHOLE	16.82	0.2030	0.2049
10 / NeST-Com	17.41	0.1922	0.1837
10 / Ada-TokenCom	13.16	0.2367	0.2582
NeST–WHOLE ΔPSNR: +0.592 [0.544, 0.643] dB			
19 / WHOLE	18.64	0.1718	0.1458
19 / NeST-Com	19.63	0.1554	0.1150
19 / Ada-TokenCom	17.69	0.1496	0.1078
NeST–WHOLE ΔPSNR: +0.994 [0.940, 1.048] dB			

At $N = 512$, all five NeST–WHOLE intervals favor NeST at each SNR, with PSNR gains of 0.50–0.99 dB. At 10 dB both use target $(7, 0)$ and 16-QAM rate 2/3; utilization rises from 79.30% to 99.54%. Against Ada, NeST has higher PSNR and lower LPIPS throughout, and lower DISTS/DreamSim at 4/10 dB. Ada is better on both distances at 19 dB: NeST–Ada differences are $0.0058\,[0.0036, 0.0079]$ and $0.0072\,[0.0037, 0.0106]$. Supplement S2 gives CLIP-I and rejection counts.

Without recalibration, Kodak24 center crops at $N = 1024$ give Token–WHOLE PSNR gains of 0.567/0.794/0.405 dB at 4/10/19 dB, with favorable DISTS/DreamSim paired intervals at all three points. Token also improves both distances over Quarter at 10/19 dB, while the 4-dB intervals cross zero. Against Ada, Token has lower distances at 4 dB; Ada has lower DISTS at 10/19 dB, with DreamSim intervals crossing zero. All 864 four-method events succeed. Supplement S8 reports all five metrics and 24-source paired intervals.

E. Operation Near the Channel Threshold

With 10-dB policies fixed, 100 sources and 20 noise realizations per point yield 30 000 receptions over 8–10 dB. All three methods reject every frame at 8 dB. At 9 dB, WHOLE/NeST/Ada rejection rates are 15.75/14.95/13.50%; these fall to 0.10–0.20% at 9.5 dB, with no observed rejection at 10 dB. Overall quality includes rejected-frame gray outputs; successful-only quality is in Supplement S5. Source-reconstruction gains are realized within the reliable operating range of the selected digital-link policy.

IV. Conclusion

NeST-Com uses incomplete next-scale boundaries in fixed digital frames through independently terminated prefixes, actual-length fitting and prefix-constrained reconstruction. Shared-

model and matched controls show that delivered boundary tokens improve residual reconstruction and finer-scale conditioning. Token-wise fitting retains the largest feasible source-token count up to the calibrated target, while coarser within-scale grids offer lower processing cost. The evaluated system provides favorable source-referenced reconstruction at tight budgets, with budget-dependent perceptual rankings and explicit processing-cost choices.

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